Vision-Based Settling Rate Analysis for Water Treatment

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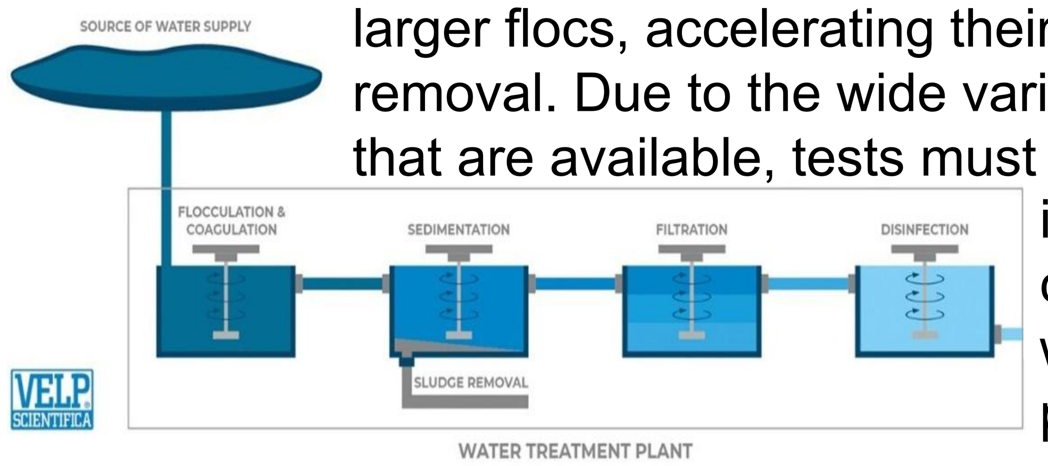
Background

Water treatment relies on the effective removal of suspended solids and other contaminants to ensure water quality and safety. The first critical step in water treatment is the addition of coagulants to bind suspended particles in raw water to form larger flocs, accelerating their settling and removal. Due to the wide variety of coagulants that are available, tests must be conducted to identify the best coagulant for each water treatment process.



Existing Solutions

Jar Tests
(Current Industry Standard)

Jar Test using ML model
(Previous Capstone Project)

Jar testing relies on manual observation which introduces variability and limits portability. There is a growing need for automated, on-site testing to improve accuracy and speed in water treatment. Our team is developing a portable, computer vision-based apparatus using machine learning to quantify settling rates, building on a previous prototype while improving mobility and enabling comparison of multiple samples.

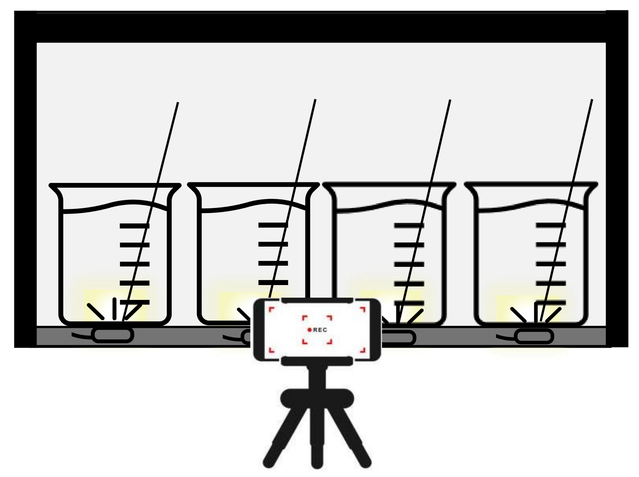
Design Specifications

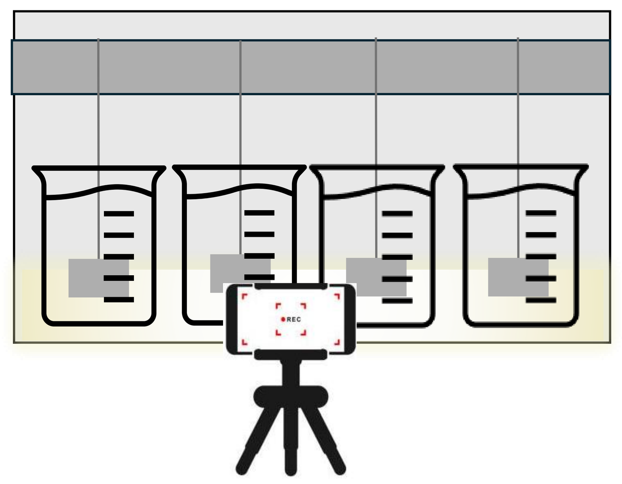
Improve how settling performance is assessed during jar tests and pilot runs

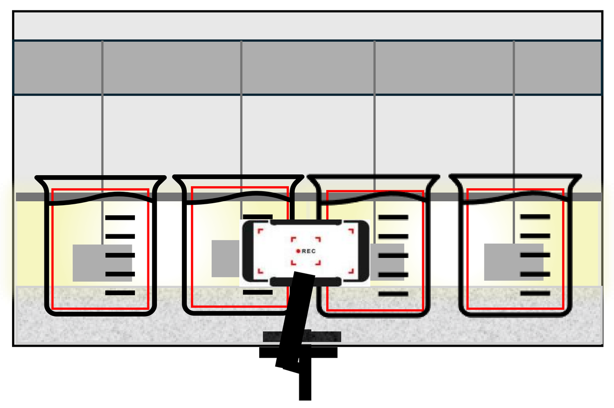
- Standardization: Make automated and objective settling measurements in real-time
- Portability: Develop a portable testing apparatus to be used on a job site for real-time results
- Speed & Efficiency: Reduce test time by increasing number of jars that can be run at once

Requirements: Model should be compatible with iPhone products

Concept Evaluation

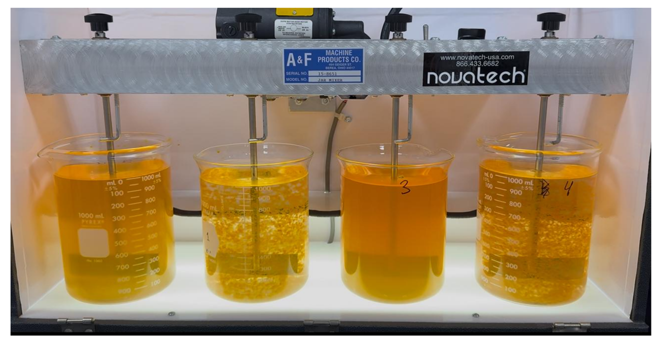






Design A	Design B	Design C
Pros • Most control over dimensions (portability) • Smaller water samples required	Pros • Most familiar • Cost effective (existing equipment) • Fewest modification and most robust	Pros • Removable modifications • Controls for the most variables • Strongest signal for mean-intensity turbidity
Cons • Least robust • Unfamiliar • Poor mixing (manual)	Cons • Lighting issues • Camera misalignment (poor for model training)	Cons • Least portable • Modified to specific apparatus
Worst	Middle	Best

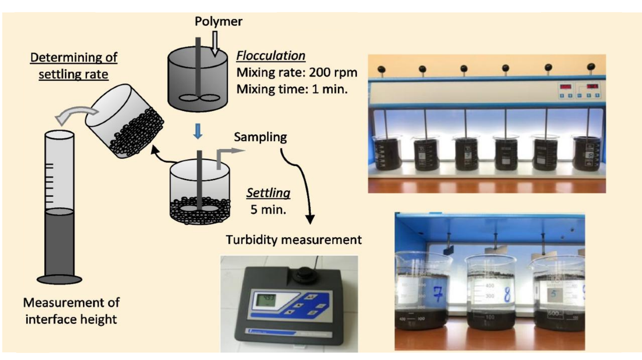
Design Methodology



Experimental
Confirm structural robustness and determine the effect of jar position on computation and create a calibration curve

Computational
- **OpenCV** - image processing
- **TensorFlow** - overall model development
- Video frame is processed through **background subtraction** and **settling curve generation** to accurately quantify the settling behavior

Validation
Complete by comparing the system's automated measurements to manual results and standard turbidity readings.



Design Details

Experimental Methods

Backlighting reduces glare and shadowing. Modifying the jar test apparatus improves durability and data capture. A fixed phone stand ensures consistent camera angles.

Computational & Validation Methods

Settling curves and clarity plots will be generated from 30 FPS video data with pixel intensity indicating settling rate. Consistent camera positioning and LED backlighting ensure reliable measurements

Design Details/Specifications

Novatech AFM-JM-4 apparatus was modified for this project
Phone stand attached to rig for consistent camera position
Back stopper was raised to allow **backlighting**
Nonslip bottom added to improve jar stability
Video captures at **30 FPS** with **auto-cropping capabilities**
Overall dimensions were **23” x 9” x 16.25”**

Cost Analysis

Equipment		Price
*Mini-Jar Apparatus		$4800
Lighting Modifications		$40
Camera Positioning Control		$30
Jars best for Model		$25
Surface Modifications		$25
Camera to computer (data collection)		$10
*assumed possession of phone camera		*$700
Total w/o Apparatus	Total w Apparatus	Total w/ Phone
$130	$4920	$5620

Design and Future Work

After evaluating multiple configurations, the fixed camera with LED back-lighting proved the most effective; offering higher accuracy, portability, and consistent results with relatively simple construction.

Future work
- Conduct jar tests to collect training data for the CNN
- Refine lighting and calibration curve to ensure differences between jar positions has minimal impact on testing results
- Tune the CNN model for higher prediction accuracy

Acknowledgements & References

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